GALOIS GROUPS IN ENUMERATIVE GEOMETRY
Spring 2025

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Abstract. https://github.com/tbrazel/galois-notes

0. About

Some notes from a mini-seminar run in Spring 2025 on Galois groups of enumerative problems, following Joe Harris' 1979 paper of the same name [Har79]. These notes aren't intended to be a definitive reference, but are perhaps more narrow than the survey paper [SY21], for instance. Our goals are to fill out details and examples to better understand how to carry out computations in monodromy and enumerative geometry.

0.1. **Notation.** We modify some notation slightly (e.g. $\mathrm{Gr}(k,n)$ instead of $G(k,n)$ for the Grassmannian of affine k-planes in affine n-space), but for the most part follow closely with the notation in [Har79].

1. Talk 1: Thomas Brazelton, 3/10

Question 1.1. Given an enumerative problem, to what field extension do we need to pass to in order to find all its solutions?

This question has its roots in 19th century algebraic geometry, and was crystallized by Jordan in Chapter III of his treatise on Galois theory:

> *Les solutions des problèmes dont il s'agit sont en général assez nombreuses, mais liées les unes aux autres par certaines relations géometriques. De ces relations on déduit immédiatement l'existence d'une fonction entière $\phi(x_0, x_1, \ldots)$ dont le group contient celui de l'équation X. Réciproquement, si l'on était certain de connaître* toutes *les relations géométriques que présente la question proposée (ou du moins celles dont les autres dérivent), le group de l'équation X contiendrait toutes les substitutions du group de $\phi(x_0, x_1, \ldots)$. Mais une semblable certitude est difficile à obtenir, malgré le soin apporté par d'habiles géomètres à l'étude de ces problèmes. Il ne serait donc pas impossible que les équations auxquelles ces problèmes donnent naissance eussent parfois une forme plus particulière encore que celle que nous allons trouver, en nous appuyant sur les résultats obtenus par nos prédécesseurs*
>
> Camille Jordan, 1870, [Jor70, pp.301-302]

Question 1.2. Are these sorts of questions *solvable*? Meaning solvable in radicals?

The six subsections of Chapter III in Jordan's book are dedicated to particular problems whose Galois group he is interested in studying. For instance:

III.I (Section 2) Given a planar cubic curve, we know by work of Hesse that it has the property that the line passing through any two of its flexes pass through the cubic at a third inflection point. As there are nine flexes, each flex lies on four of these lines, for a total of 12 lines.[1] *Can we obtain a formula for the flexes in terms of the coefficients of the cubic?*

III.II Given a quartic curve, can we find a cubic curve so that among the 12 points of intersection, they are in three sets of four colinear points? This follows work of Clebsch, who asked *contact problems* of a similar flavor (see [Jor70, (429)]).

III.III Again following work of Clebsch, given a quartic surface with a double conic, we can find five cones whose edges are bitangent to the surface. There are 16 lines on the quartic surface, each of which intersect the doubled conic at a single point. Can we solve for the lines given the quartic surface or the cones?

III.IV Kummer showed there exist quartic surfaces with 16 singular points, which lie in groups of six on singular tangent planes, each of which intersect other tangent planes at these six points. Can we solve for these points (or these planes) given the Kummer surface?[2]

III.V Following Steiner,[3] every smooth cubic surface has 27 lines. These lines lie on 45 tritangents, and two tritangent planes always intersect at some line, not necessarily a line on the cubic surface though. If two tritangent planes $a_1b_1c_1$ and $a_2b_2c_2$ don't meet at a line on a cubic surface, then there exists another tritangent plane $a_3b_3c_3$ for which $a_1a_2a_3$, $b_1b_2b_3$, and $c_1c_2c_3$ form tritangents. These are called *trihedral pairs*, and there are 120 of them (see e.g. [Hun96, p. 112]). Can we solve for the equations of the lines given the equation of the cubic surface? Can we incorporate the constraints imposed by the trihedral pairs?

III.VI Again following Clebsch, fix a curve C of order n and $\frac{n(n-3)}{2}$ points on it. Can we determine the curves of order $n-3$ intersecting C? For example if $n = 4$, can we determine all the 28 bitangents to C?

The idea of a *Galois group* of an enumerative problem is a setting in which we can capture these kinds of algebraic questions. The right way to phrase these kinds of questions is in the language of generically finite maps of varieties.

1.1. **Setup and goal.** Let X and Y be irreducible algebraic varieties of the same dimension over $\mathbb{C}$, let $\pi\colon Y \to X$ have degree $d > 0$. This induces a map on function fields $K(X) \to K(Y)$ which is a finite field extension.[4] and necessarily separable since we are in characteristic zero.

We're going to pick some nice $p \in X$ (in the region over which π is finite), and look at its fibers $\pi^{-1}(p) = \{q_1, \ldots, q_d\}$. We'll define two ways to permute the fibers – one coming from Galois theory and one coming from monodromy, and we'll demonstrate that these are equal, following [Har79, §1].

1.2. **The Galois group, formally.** By the primitive element theorem, $K(Y)$ is generated over $K(X)$ by some rational function $f \in K(Y)$ which satisfies a minimal polynomial relation:

$$P(f) = f^d + g_1 f^{d-1} + \ldots + g_{d-1} f + g_d = 0,$$

where $g_1, \ldots, g_d \in K(X)$.

[1] I'd love to include a nice picture here, but unfortunately by a result of Klein, a smooth real planar cubic will have at most three of its nine flexes defined over $\mathbb{R}$ [Ron98].

[2] This configuration admits a really nice contemporary description in terms of θ-characteristics. We should come back and fill out the details here.

[3] The Steiner reference is his 1857 paper *Über die Flächen dritten Grades* (On cubic surfaces) [Ste57], published in Crelle's Journal, which was called Borchardt's Journal during Borchardt's tenure as editor (1856–1880). It's interesting to me that Jordan references work of Steiner, rather than work of Cayley and Salmon, from 1849 and 1847, respectively.

[4] If X and Y are affine, it's clear this function field extension is finite. In the general case, we can reduce to the affine case by looking at the generic points (since they're irreducible), see e.g. [Aut, 02NW].

Recall that $\mathcal{O}_X$ is the sheaf of holomorphic functions. We obtain the sheaf $\mathcal{K}_X$ of meromorphic functions as the quotient ring. We get an injection of sheaves $\mathcal{O}_X \to \mathcal{K}_X$. In the general scheme-theoretic setup, this need not be a field, however since our X is particularly nice (integral, Noetherian,..) it will be. So we can look at the germs of meromorphic functions around p, which forms a *field* $\mathcal{K}_{X,p}$. The covering map induces an isomorphism of fields at each q_α:

$$\pi_\alpha := \pi_* \colon K_{Y,q_\alpha} \xrightarrow{\sim} K_{X,p}.$$

Let's define ϕ to be the inclusion of fields obtained by restricting global meromorphic functions around p:

$$\phi \colon K(X) \hookrightarrow \mathcal{K}_{X,p},$$

and ϕ_α to be the composite

$$K(Y) \to \mathcal{K}_{Y,q_\alpha} \xrightarrow{\pi_\alpha} \mathcal{K}_{X,p}.$$

We can sit everything inside $\mathcal{K}_{X,p}$, so let's fix some notation:

(1) K is the image of $K(X)$ in $\mathcal{K}_{X,p}$ – it is the restriction of global meromorphic functions on X to a neighborhood of p
(2) L is the subfield of $\mathcal{K}_{X,p}$ generated by all the images of the ϕ_α's. — this is meromorphic functions around p, which are coming from meromorphic functions on Y that have been restricted to some neighborhood of some q_α

Finally we take our elements $f \in K(Y)$ and $g_i \in K(X)$ and sit them inside the larger field:

$$\begin{aligned} \phi \colon K(X) &\to K \subseteq \mathcal{K}_{X,p} \\ g_i &\mapsto \widetilde{g}_i. \end{aligned}$$

and

$$\begin{aligned} \phi_\alpha \colon K(Y) &\to \mathcal{K}_{X,p} \\ f &\mapsto \widetilde{f}_\alpha. \end{aligned}$$

Observe that all the $\widetilde{f}_\alpha$'s are distinct! This is because in order for f to generate $K(Y)$ over $K(X)$, it must take different values at each q_α.

Moreover, each of the $\widetilde{f}_\alpha$'s satisfy the image of the polynomial relation $P(f) = 0$ in L:

$$\widetilde{P}(\widetilde{f}_\alpha) = \widetilde{f}_\alpha^d + \widetilde{g}_1 \widetilde{f}_\alpha^{d-1} + \ldots + \widetilde{g}_d = 0.$$

Let's think about $\widetilde{P}$ as a polynomial in $L[t]$. It is a degree d polynomial with d distinct roots, given by the $\widetilde{f}_\alpha$'s.

We claim then that L is the normalization of $K(Y)/K(X)$, (which is identically K_α/K. This is because the minimal polynomial $P(t)$ splits in L, and L is the minimal field over which this occurs. So we have argued:

Proposition 1.3. $\mathrm{Gal}(L/K)$ is Galois.

The action of the Galois group permutes all the $\widetilde{f}_\alpha$'s, which permutes the indices α, giving us an inclusion

$$\mathrm{Gal}(L/K) \hookrightarrow \Sigma_d.$$

The image of this is the *Galois group* of the enumerative problem.

1.3. **The monodromy group.** Since $\pi \colon Y \to X$ is an branched cover, it satisfies homotopy lifting away from the branched points. That is, any path in X which doesn't pass through the branch locus can be lifted to a path in Y after a starting point has been chosen.

To that end, pick some $U \subseteq X$ Zariski open containing our p and $V = \pi^{-1}(U)$ so that $\pi\colon V \to U$ is an unbranched cover. Then we obtain an inclusion

$$\pi_1(U, p) \to \Sigma_d,$$

given by the action of the deck group. The image of this is called the *monodromy group* of our enumerative problem.

1.4. **Monodromy and analytic continuation.** Let X be a Riemann surface, and consider a path $\gamma\colon [0,1] \to X$, and let's take two holomorphic germs $f \in \mathcal{O}_{X,\gamma(0)}$ and $g \in \mathcal{O}_{X,\gamma(1)}$. We say g is the *analytic continuation of f along γ* if there is a finite sequence of open sets U_i along the image of γ[5] and functions $f_i \in \mathcal{O}(U_i)$ so that $f_1 = f$ and $f_n = g$.

Theorem 1.4. If $\pi\colon Y \to X$ is an unbranched cover of a Riemann surface X, and $\gamma \in \pi_1(X, x)$ is some loop, then for any $f \in \mathcal{O}_{X,x}$, any choice of lift $\widetilde{\gamma}\colon [0,1] \to Y$, and any germ $g \in \mathcal{O}_{Y,\widetilde{\gamma}(0)}$ with $\pi_* g = f$, we have that analytic continuation of g along $\widetilde{\gamma}$ exists, and the resulting germ $\widetilde{g}$ also satisfies $\pi_* \widetilde{g} = f$.

1.5. **The main result.** We are going to prove the following:

Proposition 1.5. The Galois group G equals the monodromy group M in our setup.

The first step is to argue that $M \subseteq G$. That is, any permutation coming from monodromy can be realized by an automorphism of L over K. This follows via analytic continuation!

If we pick some $\gamma \in \pi_1(U, p)$, then any lift of γ to V will send some $\widetilde{f}_\alpha$ to some $\widetilde{f}_\beta$. In particular since $K = \mathrm{im}(K(X) \hookrightarrow \mathcal{K}_{X,p})$ is fixed under analytic continuation, and since any element in L is polynomial in germs at the q_α's, analytic continuation induces a field automorphism of L/K permuting the $\widetilde{f}_\alpha$'s. That is, this permutation lies in the Galois group.

For the other direction, we want to see that the containment $M \subseteq G$ is not strict. Indeed if it were, then the M-fixed subfield L^M would not be equal to K. So it suffices to argue that everything in L fixed by the monodromy action is actually in K. As we have seen, the monodromy action occurs via analytic continuation.

So take some $h \in L \subseteq \mathcal{K}_{X,p}$, and suppose h is fixed under analytic continuation along any lift of an element in $\pi_1(U, p)$. We want to argue that h is the restriction of a global meromorphic function on X to a neighborhood of p. We'll define a candidate one – define $\widetilde{h}$ on U by picking, for every $p' \in U$, an arc from p to p' and analytically continuing h along it. This is well-defined precisely because h is fixed under the monodromy action, so we obtain a well-defined value in a neighborhood of p', independent of the path we chose.

We now know that h extends to a meromorphic function $\widetilde{h}$ on U. We want to see that this extends to all of X. In order to do this, we exploit that $h \in L$. So h can be written as some polynomial in $\widetilde{h}_\alpha$'s, where $\widetilde{h}_\alpha$ is a meromorphic function on all of Y restricted to a neighborhood of q_α. None of these have essential singularities, and this is unchanged when taking a polynomial in them. Hence $\widetilde{h}$ has no essential singularities, and therefore extends to a meromorphic function on X whose germ at p is h. Thus $L^M = K$, and we are done.

1.6. **Galois groups over the rationals.** A priori this tells us nothing about Galois groups of enumerative problems over $\mathbb{Q}$, or field extensions of solutions, or anything like that. We can translate information happening over this complex function field to statements about rationals via *Hilbert irreducibility*. As a jumping off point, let's talk about monic univariate polynomials and their Galois groups (which is the origin of this story).

[5] We may assume finite since the image of γ is compact.

Example 1.6. Let $f(x) = x^n + a_1x^{n-1} + \ldots + a_n$ be the generic monic univariate polynomial of degree n, and let's think about it as living in $\mathbb{C}(a_1, \ldots, a_n)[x]$. Then there exist *rational numbers* $a_1, \ldots, a_n$ for which $f(x) = x^n + a_1x^{n-1} + \ldots + a_n$ is irreducible in $\mathbb{Q}[x]$. In fact the set of points $(a_1, \ldots, a_n) \in \mathbb{A}^n_{\mathbb{Q}}$ for which this property holds is Zariski dense.

Corollary 1.7. If G is the Galois group of $\mathbb{C}(a_1, \ldots, a_n)[x]/f$ then G is the Galois group of any specialization of f to $\mathbb{Q}[x]$ which is irreducible. In particular, we can compute in the generic case that $G = S_n$.

This admits a generalization to polynomials in arbitrarily many variables, so we get an application which we will refer to as Hilbert irreducibility in these notes.

Slogan 1.8 (Hilbert irreducibility)**.** Let X be a complex moduli space of geometric objects (e.g. cubic surfaces), and $\pi\colon Y \to X$ a generically finite map. Then the Galois group of π is isomorphic to the Galois group of a general object in X defined over the rationals.

As a particular case to illustrate what we mean, we will see that the Galois group of lines on complex cubic surfaces is $W(E_6)$. That implies that if X is any randomly chosen cubic surface defined over $\mathbb{Q}$, the field of definition $K/\mathbb{Q}$ of the lines on X has Galois group $\mathrm{Gal}(K/\mathbb{Q}) \cong W(E_6)$.

1.7. **An example we know and love.** Consider the incidence variety of roots of a univariate polynomial

$$Y = \{(f, t)\colon f(t) = 0\} \subseteq \mathbb{P}^d \times \mathbb{P}^1.$$

Then $Y \to \mathbb{P}^d$ is a projective bundle, whose Galois group is the full symmetric group. There are many ways to prove this, we highlight two:

(1) We can argue algebraically that the Galois group of the generic polynomial of degree d is the full symmetric group Σ_d.
(2) We can argue that the monodromy of the cover is symmetric, e.g. by showing it is 2-transitive and contains a transposition.

In either of these approaches, we obtain the Abel–Ruffini theorem as an immediate application via Hilbert irreducibility.

Theorem 1.9. If $f(x) \in \mathbb{Q}[x]$ is a polynomial of degree ≥ 5, there is no general formula in radicals for the roots of f in terms of the coefficients of f.

2. Flexes and bitangents

2.1. **Historical background.** The study of high degree plane curves (degrees three or greater) dates back to Newton, but one of the first major results in this are came from Plücker (*System der analytischen Geometrie*, page 264).

Theorem 2.1 (Plucker, 1835)**.** A general degree d planar curve has $3d(d-2)$ inflection points.

Plücker proved this by equation-bashing. We provide a slightly differeent proof, due to Hesse but likely predating him (see [Gra10, p. 169]).

Hesse's proof. Recall that the radius of curvature of a planar curve $F(x, y, z) = 0$ at a point (x_0, y_0, z_0) is the reciprocal of $\det(HF)_{|(x_0,y_0,z_0)}$, where HF is the Hessian matrix of F. Note that the equation $\det HF = 0$ is homogeneous of degree $3(d-2)$, and note further that a flex on a curve is precisely a point with infinite curvature radius. Hence we can count inflection points on F via the intersection with its Hessian curve, hence by Bézout's theorem we have $3d(d-2)$ flexes. $\square$

The so-called *Plücker formulas* are highly related, and come from the same text. Plücker computed what is now called the *class* of the curve C, namely the number of tangent lines to C through another point p on the plane. Note that under pole-polar duality, this is also the degree of the *dual curve*. He observed by direct computation that the class of a general curve of degree d is $d(d-1)$.

In this direct algebraic approach, any line through p passing through two points of C (counted with multiplicity) qualifies as a tangent. Presuming that C is smooth (and if we take C to be general this is the case) then this is true on the nose, however even if C is mildly singular this fails. Plücker observed that the quantity $d(d-1)$ overcounts the honest class of the curve by two for every double point of C and by three for every cusp of C. This leads us to the formula

$$d^* = d(d-1) - 2\delta - 3\kappa$$

where δ is the number of nodes of C and κ is the number of cusps. This is the *Plücker formula*, and it resolves the so-called "duality paradox" which plagued the study of pole-polar duality since its invention. A nice exposition to these results is in Coolidge's treatise [Coo59, Chapter VI].

Terminology 2.2. A reduced irreducible planar curve C is said to be a *Plücker curve* if it falls under the scope of the Plücker formula – explicitly, if the only singularities of C and C^* are cusps and simple nodes.

Remark 2.3.

(1) More general Plücker formulae hold in which singularities are allowed to be much more badly behaved.
(2) Klein proved analogues of the Plücker formulae for real curves, treating split and non-split nodes differently. Leveraging his formula we are able to prove new results and give contemporary proofs of classical results known to Plücker, such as the fact that at most three of the nine flexes on a real cubic are real.
(3) Plücker's arguments were fairly nonrigorous, and it is ahistorical to attribute to him rigorous arguments that appear to be easy applications of his formula (for instance the computation that there are 28 bitangents to a planar quartic, attributed either to Hesse in 1848 or Jacobi in 1850).

Under duality, the Plücker formulae allow us to relate the numbers of bitangents and the numbers of flexes, since a bitangent to C is a node on C^*, and a flex on C is a cusp on C^*. If C is a Plücker curve of degree d, we then have the relationship

$$d = d^*(d^*-1) - 2b - 2f.$$

Combining this with the number $3d(d-2)$ of flexes on C, we get

$$d = d^*(d^*-1) - 2b - 2 \cdot 3d(d-2).$$

Let's now suppose that C is suitably general, so that it is nonsingular and hence $d^* = d(d-1)$ with no correction term. We then get

$$d = d(d-1)\left(d(d-1)-1\right) - 2b - 3 \cdot 3d(d-2).$$

Solving for b we obtain

$$b = \frac{1}{2}d(d-2)(d^2-9). \tag{2.1}$$

This is the number of bitangents on a general smooth curve of degree d as estimated by Plücker and proven by Jacobi.

2.2. **The Plücker formulas via Riemann-Hurwitz.** TODO

2.3. **Monodromy of flexes: the setup.** Let $W_d := \mathbb{P}(H^0(\mathcal{O}_{\mathbb{P}^2}(d))) \cong \mathbb{P}^{\binom{d}{2}-1}$ be the complete linear system of degree d plane curves, and let I_0 be the locus of points on lines

$$I_0 := \{(p, \ell) \colon p \in \ell\} \subseteq \mathbb{P}^2 \times (\mathbb{P}^2)^*.$$

We let

$$I_d := \{(C, p, \ell) : m_p(C \cdot \ell) \geq 3\} \subseteq W_d \times I_0$$

be the incidence variety of curves equipped with a flex. We obtain projection maps

$$\begin{tikzcd} & I_d \arrow[dl, "\pi"'] \arrow[dr, "\eta"] & \\ W_d & & I_0. \end{tikzcd}$$

Proposition 2.4. If $d \geq 3$, then π is generically finite of degree $3d(d-2)$.

Proof. We can argue that the general curve of degree d has only flexes and bitangents (it doesn't admit a hyperflex, a tritangent, or a flex bitangent). Hence over this open locus, π has a well-defined degree. Note we restrict to $d \geq 3$, since a conic has no flexes. Obtaining the degree is exactly the Plücker formula, which as we have seen can be derived in multiple different ways. □

We want to try to find the monodromy group of π, and the following fact will be helpful.

Proposition 2.5. The incidence variety I_d is irreducible.

Proof. We can consider the projection $\eta \colon I_d \to I_0$. Since I_0 is irreducible, we will be able to conclude (by Proposition A.3) that I_d is irreducible if η has equidimensional fibers. Indeed we can check that the fiber of η over any point $(p, \ell) \in I_0$ is exactly those degree d plane curves with a flex line ℓ at the point p. This is always a codimension three linear subspace of W_d. □

We now want to argue that the monodromy group $\mathrm{Mon}(\pi)$ acts transitively on the fibers of π. By covering space theory, it will suffice to argue that, by restricting to the locus $U \subseteq W_d$ over which π is unramified, the fiber $\pi^{-1}(U)$ is connected.

Proposition 2.6. The monodromy group acts transitively on the fibers of π.

Proof. todo □

Proposition 2.7. The monodromy group is 2-transitive on the fibers of π.

Proof. todo □

To argue that the monodromy group is full symmetric, it suffices to exhibit a transposition. Here we use this key lemma:

Lemma 2.8 ([Har79, p. 698])**.** Let $\pi \colon Y \to X$ be holomorphic of degree n, and suppose there exists some $p \in X$ so that $\pi^{-1}(p) = \{q_1, \ldots, q_{n-1}\}$ has $(n-1)$ distinct points, so that π is simple at $q_1, \ldots, q_{n-2}$, and π has a double point at q_{n-1}. Suppose further that Y is locally irreducible at q_{n-1}. Then the monodromy group of π contains a simple transposition, obtained by taking a small loop around p.

Proof. todo □

So we'd like to locate a planar curve with $3d(d-2)-2$ simple flexes, and exactly one *hyperflex*. Note at the hyperflex the tangent line will meet to order ≥ 4, and in particular this would violate Bézout's theorem if $d \leq 3$.

Theorem 2.9. For $d \geq 4$, the monodromy group of $I_d \xrightarrow{\pi} W_d$ is the full symmetric group $\Sigma_{3d(d-2)}$.

Proof. Fix a point $(p_0, \ell_0) \in I_0$, and consider the linear system of degree d curves with a hyperflex at (p_0, ℓ_0):

$$W'' = \{C \in W \colon m_{p_0}(C \cdot \ell_0) \geq 4\} \subseteq W.$$

A general $C \in W''$ is smooth at p_0 with ℓ_0 a simple hyperflex (the multiplicity is exactly four). How do we know that the general element in W'' has only simple flexes at the other flex points though? Consider the incidence correspondence

$$I'' = \{(C, p, \ell) \colon m_p(C \cdot \ell) \geq 3,\ p \neq p_0,\ \ell \neq \ell_0\} \subseteq W'' \times I_0.$$

The fibers of $\eta \colon I'' \to I_0$ are linear of codimension three in W'', therefore I'' is irreducible of dimension $\dim I'' = \dim W''$ (todo: why?). Then... □

2.4. **Flexes on a plane cubic: geometric properties.** The following observation is classical, and its origins are hard to trace, but it was well-known at the time of Jordan (see [Jor70, p. 302]). The configuration of lines and flexes along a planar cubic is called the *Hesse arrangement* (see [Dol12, p. 118]).

Proposition 2.10. Let C be a smooth planar cubic, and consider its nine inflection points.

(1) These nine points lie on twelve lines
(2) The twelve lines meet four at a time at any given flex.

This configuration is illustrated in Figure 1.

Proof. This can be seen by endowing C with an abelian structure as an elliptic curve, after which the flexes of C can be identified with the 3-torsion points in the elliptic curve structure. □

Jordan noted (in modern language) that any monodromy of the flexes over the moduli of cubics will preserve the Hesse arrangement, in particular the resulting permutation of the nine flexes will preserve the colinearity properties they satisfy. He proved the following result.

Theorem 2.11 ([Jor70, Theorem 425])**.** Let $f(x)$ be an irreducible polynomial of degree nine, and let $\psi(x, y)$ be a rational function of two variables, which is symmetric in the variables. Suppose that f has the property that, given any two of its roots a and b, we can define a third root by the formula $c = \psi(a, b)$, which further satisfies:

$$b = \psi(a, c) \text{ and } a = \psi(b, c).$$

Then the Galois group of f is contained in the affine special linear group $\mathrm{ASL}(2, \mathbb{F}_3)$.

The immediate application of this is of course:

Corollary 2.12. The monodromy of flexes on smooth plane cubics is contained in $\mathrm{ASL}(2, \mathbb{F}_3)$.

Jordan didn't argue this was an equality, although Dickson and Weber did shortly thereafter (modulo a technicality about which field we are working over – a technicality which is explained by Hilbert irreducibility and the fact that nine is odd. We should explain further). We will prove this is an equality following [Har79, § II.2].

Question 2.13. Can we discuss Jordan's theorem in the language of resolvent degree?

Note 2.14. This group $\mathrm{ASL}(2, \mathbb{F}_3)$ acts on the cubic via projective transformations, and hence forms a subgroup of $\mathrm{PGL}_3(\mathbb{C})$ often called the *Hesse group* [Dol12, §3.1.4].

Note 2.15. This group $\mathrm{ASL}(2, \mathbb{F}_3)$ has GAP ID `[216,153]`.

Question 2.16. Does this group (as so many other groups in this field) admit any exceptional isomorphisms which hint at other ways to visualize the monodoromy?

2.5. **Monodromy of flexes on plane cubics.**

The Jacobian and Abel's Theorem. Let C be a smooth curve of genus g over $\mathbb{C}$. We have $H_1(C;\mathbb{Z}) \cong \mathbb{Z}^{2g}$ and by Serre duality, $H^0(C, \omega_C) \cong H^1(C, \mathcal{O}_C) \cong \mathbb{C}^g$. There is a natural inclusion $i : H_1(C;\mathbb{Z}) \to H^0(C, \omega_C)^\vee$ sending the path $(p \to q)$ to the functional $\int_p^q$. The *Jacobian* of C is

$$J(C) := \frac{H^0(C, \omega_C)^\vee}{H_1(C;\mathbb{Z})} \cong \frac{\mathbb{C}^g}{\mathbb{Z}^{2g}}.$$

If we choose a base point $p_0 \in C$, then for all $d \geq 1$, the *Abel-Jacobi map* $\mu_d : C^d \to J(C)$ sends a d-uple $(q_1, \ldots, q_d)$ to the functional $\left[\sum_{i=1}^d \int_{p_0}^{q_i}\right]$.

Theorem 2.17 (Abel's Theorem cite)**.** Two divisors D_1 and D_2 of degree d on C are linearly equivalent if and only if their images under the Abel-Jacobi map μ_d are equal. Moreover, the Abel-Jacobi map factors as

$$C^d \to \mathrm{Pic}_d(C) \xrightarrow{-d\cdot p_0} \mathrm{Pic}_0(C) \xrightarrow{\cong} J(C).$$

Corollary 2.18. The number of n-torsion points in $\mathrm{Pic}_0(C)$ is n^{2g}.

Proof. The group $\mathrm{Pic}_0(C)$ is isomorphic to the Jacobian $J(C) \cong (S^1)^{2g}$. The subgroup of n-torsion points is then $(\mathbb{Z}/n)^{2g} \subset (S^1)^{2g}$ and has order n^{2g}. □

Proposition 2.19. For an elliptic curve (E, o), the Abel-Jacobi map $\mu_1 : E \to J(E)$ is an isomorphism.

Flexes of a Plane Cubic and Monodromy. Let (E, o) be an elliptic curve. The complete linear series $|3o|$ embeds E as a smooth plane cubic curve. The divisor $3o$ is then linearly equivalent to the intersection of E with a general line ℓ.

If $p \in E$ is a flex point with tangent line ℓ, then $3p \sim 3o$, which means $[3(p - o)] = 0$ in $\mathrm{Pic}_0(C)$. By Corollary 2.18, there are $3^2 = 9$ points in $\mathrm{Pic}_0(C)$ that are 3-torsion, so E has 9 flexes.

Recall that $E \cong J(E) \cong \mathbb{C}/\Lambda$ where the lattice $\Lambda = H_1(E;\mathbb{Z})$. If $z \in \mathbb{C}$ is such that $z($ mod $\Lambda)$ is a flex, then $z - o \in \frac{1}{3}\Lambda$. Therefore, the flexes of E can be obtained as the cokernel $\operatorname{coker}\left(\Lambda \xrightarrow{\cdot 3} \Lambda\right) \cong \mathbb{A}^2_{\mathbb{F}_3}$. Upon choosing an origin for $\mathbb{A}^2_{\mathbb{F}_3}$, the intersection form on $\Lambda = H_1(E;\mathbb{Z})$ descends to a bilinear form $\alpha : \mathbb{F}_3^2 \times \mathbb{F}_3^2 \to \mathbb{F}_3$.

Proposition 2.20. The monodromy group $\mathrm{Mon}(\pi)$ preserves the structure of the affine space $\mathbb{A}^2_{\mathbb{F}_3}$. Upon choosing an origin $O \in \mathbb{A}^2_{\mathbb{F}_3}$, the stabilizer subgroup Stab_O preserves the $\mathbb{F}_3$-bilinear form α.

Proof. Todo. □

2.6. **Solvability of flexes in radicals, in the style of Jordan.** Observe that $\mathrm{ASL}_2(\mathbb{F}_3)$ is a *solvable group* – this was first noticed by Jordan in [Jor70, III.III§1]. An example subnormal series

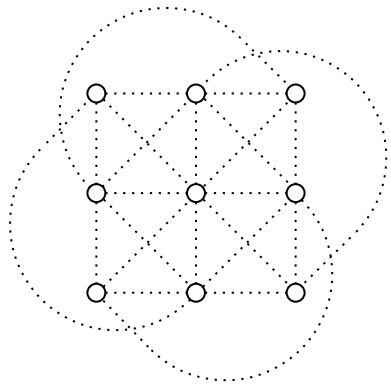

FIGURE 1. Lines on affine plane over $\mathbb{F}_3$.

exhibiting solvability is:

$$\begin{tikzcd} C_4 \arrow[r, hook] & Q_8 \arrow[r, hook] \arrow[d, two heads] & \mathrm{SL}_2(\mathbb{F}_3) \arrow[r, hook] \arrow[d, two heads] & \mathrm{ASL}_2(\mathbb{F}_3) \arrow[d, two heads] \\ & C_2 & C_2 & \mathbb{F}_3^2. \end{tikzcd}$$

Here for instance

$$C_4 = \left\langle \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \right\rangle \leq \left\langle \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} \right\rangle = Q_8.$$

In particular this solvability implies that the equation for a flex point on a cubic admits a formula in radicals in terms of the coefficients for the cubic.

Question 2.21. Can we find this formula written out explicitly anywhere?

2.7. **Solving for flexes on a given cubic.** A textbook treatment is described on [MBD16, p. 333], in a section appropriately entitled *Group G of the Equation X for the Abscissas*[6] *of the Points of Inflexion*. For a general cubic $f(x, y, z) = 0$, we can restrict our attention to the affine patch $z = 1$. We then set up the equation for $f(x, y, 1) = \det Hf = 0$, and via elimination theory we can write y as a rational function in x:

$$y = \phi(x) \in \mathbb{C}(a_0, a_1, \ldots, a_9)(x),$$

where the a_i's are the coordinates of a general cubic. Finally we plug in $f(x, \phi(x), 1) = 0$ and clear denominators to obtain a degree nine polynomial in x, the roots of which are the x-coordinates of all the nine flexes on the cubic.

3. Monodromy of bitangents

A classical problem is to compute the monodromy group of the 28 bitangents to a smooth planar quartic. As mentioned, this was a particular application for Galois theory envisioned by Jordan [Jor70, III.VI].

3.1. **History of bitangents to quartics.** As discussed earlier, the expression (Equation (2.1)) for the number of bitangents to a smooth curve of degree d, in terms of d, was known to Plücker in the 1830's. Despite this, Plücker's arguments are considered from a modern perspective (and likely also at the time) as sketchy at best. The so-called Plücker formulas were not rigorously proven by

[6]This is terminology in some older math books: *abscissa* means x-coordinate, and *ordinate* means y-coordinate.

Plücker, and although he did sketch an accurate resolution of the pole-polar duality paradox, it did not constitute a proof.

The expression in Equation (2.1) was proven by Jacobi [Jac50], an application of which is the computation that when $d = 4$, a smooth planar quartic has 28 bitangents. This was extended shortly afterwards by Hesse, a student of Jacobi, who showed that a *general* quartic has 28 bitangents [Hes55]. Hesse's work was immensely difficult, and he likened it in a letter to Jacobi to Newton's work discovering the law of gravity [Gra10, p. 165].

For a very classical textbook treatment of the theory of bitangents to quartics, section 12 (§95-§105) of Weber's 1896 book on algebra is dedicated to this [Web96, §95-§105]. A contemporary treatment can be found in [Dol12, §6.1].

Jordan, inspired by work of Clebsch, initially asked to what extent equations for bitangents can be solved for in terms of the equations for a quartic, which is precisely the question of what the Galois group is. It's unclear to me who first solved for the Galois group of bitangents. We can write it in a number of different ways:

$$O_6^-(\mathbb{Z}/2) \cong \mathrm{Sp}_6(\mathbb{Z}/2) \cong W(E_7)/\{\pm 1\}.$$

The first two describe the same group (like even beyond group isomorphism they are describing the same quadratic forms in mildly different language). The latter group passes through the theory of Lie groups. Some references for this connection:

- ▷ See [OD88, Theorem 9] for the derivation of this group in terms of the theory of Cayley octads
- ▷ Manivel [Man06] discusses this, and mentions in the intro to the paper that the Galois groups for bitangents and lines to cubic surfaces were well-known at the beginning of the 1900's. He references a connection to Lie groups that passes through the theory of Del Pezzo surfaces of degrees 3 and 2. The point of Manivel's paper is cutting out this passage through Del Pezzos and drawing a direct connection to the Lie groups. I'd like to understand both of these stories.
- ▷ In [MBD16, p. 367], the Galois group of 28 bitangents is spelled out explicitly. The 7 in E_7 comes in here through the theory of Aronhold sets — that is, given an Aronhold set, we can solve for the remaining bitangents. What is this connection, and how does it relate to the theory of Cayley octads?

3.2. **θ-characteristics for quartics.** Recall that a *θ-characteristic* on a variety X is a square root of the canonical bundle ω_X. From this perspective it can equivalently be thought of as a divisor D for which $2D \sim K_X$.

Definition 3.1. We say a θ-characteristic, represented by a line bundle $\mathcal{L}$ on X is...

- ▷ *even* if $\dim H^0(X, \mathcal{L})$ is even
- ▷ *odd* if $\dim H^0(X, \mathcal{L})$ is odd

it's entirely possible that a θ-characteristic has no nonzero sections. So we also say a θ-characteristic represented by $\mathcal{L}$ is...

- ▷ *effective* if $\dim H^0(X, \mathcal{L}) > 0$
- ▷ *vanishing* or *non-effective* if $\dim H^0(X, \mathcal{L}) = 0$.

Remark 3.2 (reference needed)**.** If C is a complex smooth projective curve of genus g, then there is a bijection between the effective θ-characteristics on C and $\mathrm{Jac}(C)[2]$, the 2-torsion points in the Jacobian of C.

Example 3.3. An effective θ-characteristic on a smooth canonical curve of genus 3 is a bitangent.

Proof. The canonical class K is the hyperplane class for the canonical embedding $C \subseteq \mathbb{P}^2$, hence as a divisor it is in general the sum of four points. If D is a θ-characteristic, it has the property that

$2D \sim K$, hence if D is effective, it is a sum of two points p and q on K. We'd love to say that $2D$ is actually given by a line, since then it would be tangent to C at both p and q. This turns out to be true – it is a feature of the linear series $|K_C|$ implied by C being projectively normal.[7] □

Proposition 3.4. On a smooth canonical curve of genus $g = 3$, a θ-characteristic is odd if and only if it is effective.

Proof. How do we prove this? □

Example 3.5. If C is a smooth quartic curve, it has 28 odd (effective) θ-characteristics, corresponding to the bitangents.

What do the even θ-characteristics correspond to? It turns out they reveal quite a bit about the geometry of the quartic. We have the following maps and constructions, many of which are bijections:

$$\begin{tikzcd}
\left\{\begin{matrix}\text{quartics and even}\\ \theta\text{-characteristics } (C,\theta)\end{matrix}\right\} \arrow[r, leftrightarrow, "(a)"] \arrow[d, leftrightarrow, "(b)"', "\sim"] & \left\{\begin{matrix}\text{space sextics}\\ \text{in } \mathbb{P}^3\end{matrix}\right\} \\
\left\{\begin{matrix}\text{nets of quadrics}\\ \text{in } \mathbb{P}^3\end{matrix}\right\} \arrow[ur, "(c)"'] \arrow[r, "(d)"'] \arrow[d, "(f)"'] & \left\{\begin{matrix}\text{symmetric determinantal}\\ \text{representations of } C\end{matrix}\right\} \arrow[u, "(e)"'] \\
\left\{\begin{matrix}\text{Cayley octads}\\ \text{in } \mathbb{P}^3\end{matrix}\right\} &
\end{tikzcd}$$

(a) *Sextic space curve embeddings from even θ-characteristics*: Given a tuple (C, θ), we can consider the very ample linear series $|3\theta| = |K_C + \theta|$. This has $h^0(C, 3\theta) = 4$, hence defines an embedding $C \hookrightarrow \mathbb{P}^3$ which we can verify is of degree six.

To go back the other way, we consider the difference between the hyperplane class on C embedded in $\mathbb{P}^3$ and its hyperplane class under its canonical embedding in the plane. The difference between these is an even θ-characteristic.[8]

(b) *Even θ-characteristics and nets of quadrics*: This bijection is described in the proof of [GH04, 6.3].

(c) *Nets of quadrics to space sextics*: Given a net of quadrics, we can consider its subset of singular elements. Singular quadric surfaces are cones, and their vertices are points in $\mathbb{P}^3$. Varying over all singular quadrics in a net, the vertices sweep out a space curve of degree six. This process is called taking the *discriminant* of the net of sextics.

(d) *Nets of quadrics and symmetric determinantal representations*: Any quadric in $\mathbb{P}^3$ can be represented by a symmetric 4×4 matrix, so given a quadric net we can pick any three quadrics to form a basis, and consider the quartic form produced by the following determinant:

$$f = \det(xA + yB + zC).$$

This is a symmetric determinantal representation of a quartic. To go the other way, we produce the quadrics $Q_A(y) = y^T Ay$ and Q_B and Q_C arising from a symmetric determinantal representation of f and look at the net they span.

(e) *Space sextics from symmetric determinantal representations*: Given a determinantal representation (A, B, C) for the quartic C, we obtain an embedding $C \hookrightarrow \mathbb{P}^3$ via

$$\begin{aligned} \phi_\theta \colon C &\to \mathbb{P}^3 \\ [x : y : z] &\mapsto \ker(xA + yB + zC). \end{aligned}$$

[7] Can we explain this more? We had a discussion about definitions during Michael's talk I'd love to clarify in the notes.

[8] **Question**: Is every space sextic a genus 3 curve embedded via an even θ-characteristic?

(f) *Nets of quadrics to Cayley octads*: Given a net of quadric surfaces, its base locus is eight points in $\mathbb{P}^3$, which we call a *Cayley octad*.

3.3. **Cayley octads, formally.**

Definition 3.6 ([Cay69, p. 20])**.** Given eight points in $\mathbb{P}^3$ which are the base locus of a net of quadrics, we say they are a *Cayley octad*.[9]

Proposition 3.7 ([Dol12, 6.3.3])**.** Any Cayley octad arising from a quartic (C, θ) has the property that no three of its eight points are colinear, and no four are coplanar.

Suppose we are handed eight points in $\mathbb{P}^3$ satisfying Proposition 3.7. We can ask whether an associated quartic and even θ-characteristic can be uniquely recovered from this data, and the answer is yes. Moreover, we don't need all eight points to rebuild the quartic – we only need *seven of the eight points*.

3.4. **Duality via an even θ-characteristic.**

Recall we can send a curve to its *dual* by sending a point on a curve to its tangent line. With an even θ-characteristic in hand, we can reframe this in a slightly different way. Recall we have this embedding

$$\phi_\theta \colon C \to \mathbb{P}^3$$

which depended upon a choice of determinantal representation arising from θ. Similarly, we have an associated net of quadrics N_θ, and we can pick a basis for them, say

$$\operatorname{span}\{Q_1, Q_2, Q_3\} = N_\theta.$$

These give rise to a rational map

$$\begin{aligned} \psi_\theta \colon \mathbb{P}^3 &\dashrightarrow \mathbb{P}^2 \\ y &\mapsto [Q_1(y) : Q_2(y) : Q_3(y)]. \end{aligned}$$

This is defined everywhere except the base locus of the net (at the Cayley octad).

Proposition 3.8. The image $\operatorname{im}(\gamma)$ of the composite of ϕ_θ and ψ_θ is the dual curve C^*:

$$\begin{tikzcd} C \arrow[d, "\phi"'] \arrow[dr, "\gamma"] & \\ \mathbb{P}^3 \arrow[r, dashed, "\psi"'] & \mathbb{P}^2. \end{tikzcd}$$

This perspective makes the connection to bitangents slightly more clear. If we look at the eight points $\{p_1, \ldots, p_8\}$ forming the indeterminacy loci of ψ, the image of $\psi(\overline{p_i p_j})$ is a single point in the dual projective plane, yielding a bitangent to the original quartic. There are $\binom{8}{2}$ such lines, yielding 28 bitangents.

See for instance [KV24, §2] for a more careful treatment of this.

3.5. **Computing the monodromy of bitangents to plane curves.**

... to show the monodromy is transitive, we have to show (by covering space theory) that the covering space is connected. To do this, we consider the projection

$$\eta \colon J_d \to J_0 \cong \operatorname{Conf}_2(\mathbb{P}^2).$$

The fiber of η over a point in J_0 is a linear subspace in J_d. So η is a fiber bundle, hence J_d is irreducible, hence connected. Then the monodromy is transitive.

Proposition 3.9. The monodromy acts 2-transitively on the fiber.

[9] These arise in Cayley's work via the study of *octadic surfaces*, being quartic surfaces with eight nodes.

Proof. It suffices to argue that the stabilizer of a bitangent, as a subgroup of monodromy, acts on the remaining points in the fiber transitively.... TODO □

To exhibit a simple transposition, we want to find a curve with one fewer than the generic number of bitangents. It will be a curve with a simple flex bitangent. This will exist for $d \geq 5$ (because we can't have a flex bitangent for a degree four curve by Bézout).

3.6. **Monodromy of bitangents to quartics.** Given a bitangent $\overline{p_1 p_2}$ to a generic quartic C, note that $2p_1 + 2p_2 \sim K_C$ is the canonical class, i.e. a hyperplane section of the quartic. Conversely, since C is "nice" any divisor equivalent to K_C which has the form $2p_1 + 2p_2$ is cut out by a line. Here nice means projectively normal, i.e. canonically embedded by a complete linear series, and for every d, we have a surjection

$$H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)) \twoheadrightarrow H^0(C, \mathcal{O}_C(d)).$$

Projectively normal is normal + this condition.

Hence we get a bijection between bitangents to C and divisors $p_1 + p_2$ so that $2(p_1 + p_2) \sim K_C$.

If k is any divisor with $2k \sim K_C$, then $2(p_1 + p_2 - k) = 0$. Hence $(p_1 + p_2) - k$ is a deg 0 divisor in C and $k - (p_1 + p_2)$ has order 2.

So bitangents correspond to 2-torsion points in Pic^0.

Abel-Jacobi theorem We have that

$$\mathrm{Pic}^0(C) \xrightarrow{\sim} J(C),$$

where $J(C)$ is the Jacobian

$$J(C) = H^0(C, \Omega^1_C)^* / H_1(C; \mathbb{Z}).$$

Explicitly, if we fix a basepoint $p_0 \in C$, we send $p - p_0$ to $\omega \mapsto \int_{p_0}^{p} \omega$. A priori the target depends on the path we pick between p_0 and p, however we've modded out by $H_1(C; \mathbb{Z})$ so it doesn't matter.

Bitangents correspond to some 2-torsion points in $J(C) = \mathbb{C}^3/\mathbb{Z}^6$.[10] Here we're looking at a torus, so we have a nice structure to work with to find 2-torsion stuff.

[pic – in $\mathbb{C}/\mathbb{Z}$ there are four 2-torsion points]

We know the points of order 2 in $J(C)$ form a 6D vector space $\mathbb{F}_2^6$. This has 64 elements, so not all of them are bitangents.

How does the monodromy group act on $J(C)$? We claim it preserves the affine structure (this follows from the following: if we take four bitangents, with the property that the four lines sum to zero in the jacobian, then the eight points of tangency lie on a common conic. this geometric intersection property is preserved by monodromy), so we get that

$$\mathrm{Mon}(\pi) \leq \mathrm{AGL}_6(\mathbb{F}_2).$$

It turns out there are further restrictions.

We have a skew-symmetric non-degenerate pairing

$$\widetilde{Q} \colon H_1(C; \mathbb{Z}) \times H_1(C, \mathbb{Z}) \to \mathbb{Z},$$

given by intersection with signs. This induces a form on half of the lattice

$$4\widetilde{Q} \colon \frac{1}{2}\Lambda \times \frac{1}{2}\Lambda \to \mathbb{Z}$$
$$(v, w) \mapsto \widetilde{Q}(2v, 2w).$$

This is "manually imposing" the intersection form on the more dense lattice.

[10] Here $g = 3$ so $H^0(C, \Omega^1_C) \cong \mathbb{C}^3$.

This in turn induces a form on the quotient (where $V = \frac{1}{2}\Lambda/\Lambda$ is our 6D vector space over $\mathbb{F}_2$ as before):

$$Q\colon V \times V \to \mathbb{Z}/2.$$

It is valued in $2\mathbb{Z}$, since if we input any two things from Λ we get something even.

This is a *strictly* skew-symmetric bilinear form.

This form is preserved by $\mathrm{Mon}(\pi)$, hence

$$\mathrm{Mon}(\pi) \leq \mathrm{AO}_6(\mathbb{F}_2).$$

Associated to any symmetric bilinear form, we will get two quadratic refinements q^+ and q_-. These are helpful in determining the *effective* semicanonical divisors. That is, we can find quadratic forms satisfying $q(v) = 0$ if and only if v represents a bitangent. The monodromy group will preserve this quadratic form, so we ultimately get the monodromy group is a subgroup of the *Steiner group* H, which is $O_6(\mathbb{Z}/2)$.

Can show that $\mathrm{Stab}_{\mathrm{Mon}(\pi)}(k_0) = O_6^-(\mathbb{Z}/2)$. Sidhanth will explain!

3.7. **Why $W(E_7)$?** Given a smooth planar quartic, we can take a cover of $\mathbb{P}^2$ branched along the quartic, and we obtain a degree two del Pezzo. This is of the form $\mathrm{Bl}_7(\mathbb{P}^2)$. A line on the del Pezzo is any curve intersecting the canonical divisor with order 1, and we have 56 such lines. The image of any of these in $\mathbb{P}^2$ is precisely one of our bitangents, and they map 2-to-1. From this perspective, $W(E_7)/\{\pm 1\}$ emerges as the Galois group analogously to how $W(E_6)$ emerges as the Galois group from $\mathrm{Bl}_6\mathbb{P}^2$.

In fact, given seven points on the plane satisfying the closed condition that no three are colinear and no six lie on a conic, we obtain a nonsingular quartic together with a choice of even θ-characteristic. In fact we have a birational isomorphism

$$\mathrm{UConf}_7(\mathbb{P}^2)/\mathrm{PGL}_3 \overset{\sim}{\dashrightarrow} \mathcal{M}_3^{\mathrm{ev}},$$

where UConf is the moduli of unordered configurations, and $\mathcal{M}_3^{\mathrm{ev}}$ is the moduli of genus three curves equipped with an even θ-characteristic [Dol12, 6.3.12]

4. Lines on hypersurfaces

Question 4.1. How many lines lie on a general hypersurface $X_d \subseteq \mathbb{P}^n$ of degree d.

If $n = 3$, $d = 3$, then a smooth cubic surface has exactly 27 lines.

For $n = 4$, $d = 5$, it's not true that a *smooth* quintic threefold has the same number of lines, but a general quintic threefold has 2875 lines.

The *Fano variety* of lines parametrizes lines inside of a given projective variety $X \subseteq \mathbb{P}^n$.

Question 4.2. Before constructing this, for which n and d do we expect the existence of lines on a degree d hypersurface $X_d \subseteq \mathbb{P}^n$?

Consider the dimension of the space of all hypersurfaces

$$N = \dim |\mathcal{O}_{\mathbb{P}^n}(d)| = \binom{n+d}{n} - 1.$$

We get an incidence correspondence[11]

$$\Phi_{n,d} := \left\{(X, L) \in \mathbb{P}^N \times \mathbb{G}(1, n) : L \subseteq X\right\}.$$

[11] Here $\mathbb{G}(1, n)$ is the lines in $\mathbb{P}^n$, it is equivalently the affine Grassmannian $G(2, n+1)$.

We get two projections

$$\begin{tikzcd} & \Phi_{n,d} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ \mathbb{P}^N & & \mathbb{G}. \end{tikzcd}$$

Proposition 4.3. The incidence variety $\Phi_{n,d}$ is smooth and irreducible of dimension $\dim(\Phi_{n,d}) = N + 2(n-1) - (d+1)$.

Proof. Observe that π_2 is a projective bundle, since over any line $[L] \in \mathbb{G}$, its fiber is all those hypersurfaces containing L. This is the same as

$$\pi_2^{-1}([L]) = \mathbb{P}H^0(\mathbb{P}^n, \mathcal{I}_L(d))^\vee.$$

This comes from the ideal sheaf sequence for a line $L \subseteq \mathbb{P}^n$, tensoring with $\mathcal{O}(d)$, then taking the LES on cohomology:

$$0 \to H^0(\mathcal{I}_L(d)) \to H^0(\mathcal{O}_{\mathbb{P}^n}(d)) \twoheadrightarrow H^0(\mathcal{O}_L(d)) \to 0$$

We can show that latter map is surjective (can extend any homogeneous polynomial in two variables to one in $n+1$ variables). Since $H^0(\mathcal{O}_L(d)) = H^0(\mathcal{O}_{\mathbb{P}^1}(d))$, we have that $H^0(\mathcal{I}_L(d))$ has constant dimension.[12] Therefore π_2 is a projective bundle. We get that

$$\begin{aligned} \dim H^0(\mathcal{I}_L(d)) &= \dim H^0(\mathcal{O}_{\mathbb{P}^n}(d)) - H^0(\mathcal{O}_L(d)) \\ &= N - (d+1). \end{aligned}$$

Then we add $2(n-1)$ back in for the dimension of the Grassmannian we are working over. □

The expected dimension of lines on a general $X_d \subseteq \mathbb{P}^n$ is then

$$\dim \Phi - N = 2(n-1) - (d+1).$$

We expect to have finitely many lines when this is zero, i.e. when $d = 2n-3$.

Since we computed the dimension of Φ, the equality above is an *upper bound*.

Theorem 4.4. If the expected dimension is ≥ 0 then every hypersurface contains lines (the fiber of π_1 is nonempty for every hypersurface $X \in \mathbb{P}^N$) and the dimension of the space of lines on X, denoted $F_1(X)$, is equal to the expected dimension, for a general X.

So if we're working with general hypersurfaces, then we have the expected dimension count.

Conjecture 4.5 (Debarre-de Jong)**.** If $X_d \subseteq \mathbb{P}^n$ is *any* smooth hypersurface of degree $d \leq n$, then $F_1(X)$ has dimension equal to the expected dimension $2n-3-d$.

This is false in other degree ranges.

Example 4.6. If $n = 3$ and $d = 4$, then $X_4 \subseteq \mathbb{P}^3$ is a smooth quartic K3, and a general one doesn't have any lines. By Noether–Lefschetz the very general quartic has Picard rank one. However we can write down smooth quartics which contain lines (e.g. the Fermat quartic), which would violate the conjecture above.

The main theorem we will want to prove is

Theorem 4.7. The Galois group of lines on a general hypersurface $X \subseteq \mathbb{P}^n$ of degree $2n-3$, for $n \geq 4$, is the full symmetric group.

[12] Jake notes: if we fix a line L, then $F_{|L} = 0$ is a linear condition on F.

4.1. **Construction.** Let $L \subseteq X = \{g = 0\} \subseteq \mathbb{P}^n$. Then, thinking of $g \in H^0(\mathcal{O}_{\mathbb{P}^d}(d))$, there is a restriction map to the line L:

$$\mathrm{res}\colon H^0(\mathbb{P}^n(d)) \to H^0(\mathcal{O}_L(d)).$$

To say X contains the line is to say $g \mapsto 0$.

Key idea: If we look at the family of restrictions of sections to L, we can vary this as L ranges over points in the Grassmannian, and we can realize this as a vector bundle over $\mathbb{G}(1, n)$.

We get that the images of g under the restriction map become values of a section of this bundle.

More explicitly, let V be a vector space of dimension $n + 1$, where we are thinking of $\mathbb{P}^n = \mathbb{P}(V)$. There is a bijection between lines $L \subseteq \mathbb{P}^n$ and dimension two subspaces $\Lambda \subseteq V$. Tensoring V with $\mathcal{O}_{\mathbb{G}}$, we get a trivial bundle

$$V \otimes \mathcal{O}_{\mathbb{G}} \to \mathbb{G}.$$

Inside of this we have a tautological rank two subbundle $\mathcal{S} \subseteq V \otimes \mathcal{O}_{\mathbb{G}}$, whose fiber $\mathbb{S}_{[L]}$ is Λ, where $L = \mathbb{P}(\Lambda)$ for $\Lambda \subseteq V$ a rank two subspace.

The fiber of $\mathcal{S}^\vee$ over $[L]$ is linear forms on L, which is $H^0(\mathcal{O}_L(1))$. What does it mean for a polynomial to restrict to zero? Dualizing the inclusion $\mathcal{S} \subseteq V \otimes \mathcal{O}_{\mathbb{G}}$, we get

$$V^\vee \otimes \mathcal{O}_{\mathbb{G}} \to \mathcal{S}^\vee.$$

Note that $V^\vee \otimes \mathcal{O}_{\mathbb{G}}$ is trivial, so it's recording *constant sections*. Restricting this map to some L, we get that a linear form ϕ on V is mapped to the restricted linear form $\phi_{|L}$.

We can symmetrize this to look at degree d forms, and we get

$$H^0(\mathcal{O}_{\mathbb{P}^n}(d)) = \mathrm{Sym}^d V^\vee \to \mathrm{Sym}^d(\mathcal{S}^\vee),$$

sending $g \mapsto g_{|L}$. The constant section g gets send to some $\sigma_g \in H^0(\mathrm{Sym}^d V^\vee)$, and $F_1(X)$ is the zero locus of this section σ_g.

Proposition 4.8. The total Chern class of $\mathcal{S}^\vee$ is

$$c(\mathcal{S}^\vee) = 1 + \sigma_1 + \sigma_{1,1}.$$

We have $\sigma_g \in H^0(\mathrm{Sym}^d \mathcal{S}^\vee)$. Here $\mathcal{S}$ has rank two, so $\mathrm{Sym}^d(\mathcal{S}^\vee)$ has rank $d + 1$. Its zero locus will then be

$$\deg c_{d+1}(\mathrm{Sym}^d \mathcal{S}^\vee) = \#\{\text{lines on a general hypersurface } X_{2n-3} \subseteq \mathbb{P}^n\}$$

We use the splitting principle and get

$$dd! \sum_k \frac{(2k)!}{k!(k+1)!} \sum_{\substack{I \subseteq \{1,\dots,n-2\} \\ \#I = n-2-k}} \prod_{i \in I} \frac{(d-2i)^2}{i(d-i)}.$$

This is https://oeis.org/A027363

Remark 4.9. There is a short discussion in the paper of when $F_1(X)$ is singular. It turns out $F_1(X)$ is singular at $u \in F_1(X) \subseteq \mathbb{G}$ if and only if the corresponding line $\lambda_u \subseteq X$ has normal bundle N with the property that $h^0(N) > 2n - 3 + d$. If we remember, this is the *expected dimension*.

By Riemann–Roch, we write

$$h^0(N) = 2n - 3 + d + h^1(N),$$

so this is the same as saying $h^1(N) > 0$.

Remark 4.10 (On the Galois group)**.**

(1) We show it's transitive (follows from Φ being irreducible)
(2) We show it's 2-transitive (fix a line L and show that Stab_L is transitive. Here is where we need $n \geq 4$)

(3) Existence of a transposition: we show that there exists an $F_1(X)$ with a point of multiplicity two (also need $n \geq 4$)

Appendix A. Algebraic geometry terms and references

Here are some terms and results, laid out in the order we need to use them throughout the notes.

Definition A.1 ([Har77, p. 91])**.** If $f\colon X \to Y$ is a morphism of varieties with Y irreducible, we say it is *generically finite* if $f^{-1}(\eta)$ is a finite set, where η is the generic point in Y.

Remark A.2. If f is locally of finite type and qcqs, being generically finite admits some equivalent conditions (see e.g. [Aut, 02NW]).

Proposition A.3. Let $f\colon X \to Y$ is a map with irreducible equidimensional fibers, with Y irreducible. Suppose that either X is equidimensional or f is proper. Then X is irreducible.

Proof. A proof can be found in `https://public.websites.umich.edu/~mmustata/Note1_09.pdf` □

References

[Aut] Stacks Project Authors. *Stacks Project*. https://stacks.math.columbia.edu/.

[Cay69] Prof Cayley. "A Memoir on Quartic Surfaces". In: *Proceedings of the London Mathematical Society* 3 (1869), pp. 19–69.

[Coo59] Julian Lowell Coolidge. *A Treatise on Algebraic Plane Curves*. Dover Publications, Inc., New York, 1959.

[Dol12] Igor V Dolgachev. *Classical Algebraic Geometry: A Modern View*. Cambridge University Press, 2012.

[GH04] Benedict H. Gross and Joe Harris. "On Some Geometric Constructions Related to Theta Characteristics". In: *Contributions to Automorphic Forms, Geometry, and Number Theory*. Johns Hopkins Univ. Press, Baltimore, MD, 2004, pp. 279–311.

[Gra10] Jeremy Gray. *Worlds Out of Nothing: A Course in the History of Geometry in the 19th Century*. Springer Undergraduate Mathematics Series. London: Springer London, 2010.

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. New York, NY: Springer New York, 1977.

[Har79] Joe Harris. "Galois Groups of Enumerative Problems". In: *Duke Mathematical Journal* 46.4 (Dec. 1979).

[Hes55] Otto Hesse. "Über Determinanten und ihre Anwendung in der Geometrie, insbesondere auf Curven vierter Ordnung." In: *Journal für die reine und angewandte Mathematik* 49 (1855), pp. 243–264.

[Hun96] Bruce Hunt. *The Geometry of Some Special Arithmetic Quotients*. Lecture Notes in Mathematics 1637. Berlin ; New York: Springer, 1996.

[Jac50] C. G. J. Jacobi. "Beweis des Satzes daß eine Curve nten Grades im Allgemeinen ... (n-2)(n2-9) Doppeltangenten hat." In: *Journal für die reine und angewandte Mathematik* 40 (1850), pp. 237–260.

[Jor70] M. Camille Jordan. *Traité Des Substitutions*. Paris: Gauthier-Villars, 1870.

[KV24] Avinash Kulkarni and Sameera Vemulapalli. "On Intersections of Symmetric Determinantal Varieties and Theta Characteristics of Canonical Curves". In: *Journal of Pure and Applied Algebra* 228.5 (May 2024), p. 107538.

[Man06] L. Manivel. "Configurations of Lines and Models of Lie Algebras". In: *Journal of Algebra* 304.1 (Oct. 2006), pp. 457–486.

[MBD16] G. A. Miller, H. F. Blichfeldt, and L. E. Dickson. *Theory and Applications of Finite Groups*. Dover Publications, Inc., New York, 1916.

[OD88] David Ortland and Igor V Dolgachev. "Point sets in projective spaces and theta functions". In: *Astérisque* 165 (1988).

[Ron98] Felice Ronga. "Felix Klein's Paper on Real Flexes Vindicated". In: *Banach Center Publications* 44.1 (1998), pp. 195–210.

[Ste57] Jakob Steiner. "Über Die Flächen Dritten Grades". In: *Journal für die reine und angewandte Mathematik (Crelles Journal)* 53 (1857), pp. 133–141.

[SY21] Frank Sottile and Thomas Yahl. *Galois Groups in Enumerative Geometry and Applications*. Aug. 2021. arXiv: 2108.07905 [math].

[Web96] H Weber. *Lehrbuch Der Algebra*. Braunschweig, 1896.